

Innocent Nyalala | PhD

School of Engineering & Science, **IIT Madras Zanzibar**

P.o Box 2496 Bweleo, Zanzibar, Tanzania

 [inyalala.github.io](https://github.com/inyalala)  Email: innocent@iitmz.ac.in

Research Interest

My research focuses on designing **efficient machine learning (ML) and computer vision systems** to automate decision-making in agriculture and healthcare. By integrating **IoT, edge computing, and AI-driven image analysis**, I aim to develop scalable solutions for:

- **Smart Agriculture:** Post-harvest technologies, food processing, optimizing crop monitoring, yield prediction, and resource management to advance sustainability.
- **Medical Imaging:** Advancing prenatal diagnostics, AI driven placental analysis, morphology prediction and placental volume estimation

My long-term goal is to bridge algorithmic innovation with real-world challenges by refining **deep learning architectures, embedded systems, and sensor fusion techniques**, enabling robust, low-latency tools for resource-constrained environments.

Education

PhD in Agricultural Electrification & Automation

Nanjing Agricultural University

Nanjing, China

09/2017–12/2023

Thesis: Research on Agricultural Products Volume Estimation Method

Based on Depth Imaging and Machine Learning Technology

Supervisor: Prof. Chen Kunjie

M.Sc. in Computer Science & Technology

Nanjing Agricultural University

Nanjing, China

09/2013–07/2016

Thesis: Research and Development of an Intelligent Control and Monitoring System based on Arduino for the Greenhouse Environment

Supervisor: Prof. Xu Huanliang

Bachelor in BMIT (IT Major)

Kabarak University

Nakuru, Kenya

09/2009–08/2012

Thesis: Knowledge-based Prime Insurance Management System

Honors: *second class upper*

Work Experience

10/2024–present: Assistant Professor, **IIT Madras Zanzibar Campus, Zanzibar Tanzania**

09/2017–12/2023: Doctoral Researcher in **Nanjing Agricultural University, Nanjing, China**

01/2013–present: Technology Consultant at **NiTS Consulting, Nakuru, Kenya**

01/2013–08/2013: Technical Support Intern in **Simbanet(K) Ltd, Nakuru, Kenya**

09/2013–12/2013: Technical Support Intern in **Kenya Power Company, Nakuru, Kenya**

Honors & Awards

2017–2021: China Scholarship Council (CSC), PhD Scholarship, Nanjing Agricultural University

2020: Excellent in Both Studies and Behavior Award, for excellent performance in the 1st Semester of the academic year 2019-2020, Nanjing Agricultural University

2013–2016: China Scholarship Council (CSC), Master Scholarship, Nanjing Agricultural University

2014: Excellent International Student Leader Award, Nanjing Agricultural University

Publications

[Google Scholar Profile](#)

[ResearchGate Profile](#)

[ORCiD Profile](#)

- Zhang, Jiayu, Baohua Zhang, Zixuan Chen, **Innocent Nyalala**, Kunjie Chen, and Junfeng Gao: **A salient feature establishment tactic for cassava disease recognition, *Artificial Intelligence in Agriculture* (IF=8.2), 2024**
- **Innocent Nyalala**, Zhang Jiayu, Chen Zixuan, Chen Junlong, Kunjie Chen: **Online chicken carcass volume estimation using depth imaging and 3-D reconstruction, *Poultry Science* (IF=3.8), 2024**

- Jiayu Zhang; Baohua Zhang; Chao Qi; **Innocent Nyalala**; Peter Mecha; Kunjie Chen; Junfeng Gao: **MAIANet: Signal modulation in cassava leaf disease classification**, *Computers and Electronics in Agriculture* (IF=7.7), 2024
- Liu, Haolu, Khurram Yousaf, Zhenwei Yu, Asad Riaz, **Innocent Nyalala**, Muhammad Waqas Alam Chattha, and Kunjie Chen: **Drying process optimization of garlic slices in closed-loop heat pump drying system by Box-Behnken design**, *Journal of Food Processing and Preservation* (IF=2), 2022
- **Innocent Nyalala**, Cedric Okinda, Nelson Makange, Tchalla Korohou, Qi Chao, Luke Nyalala, Zhang Jiayu, Zuo Yi, Khurram Yousaf, Liu Chao, Chen Kunjie: **On-line weight estimation of broiler carcass and cuts by a computer vision system**, *Poultry Science* (IF=3.8), 2021
- Makange, Nelson Richard, Changying Ji, **Innocent Nyalala**, Idris Sunusi, and Samwel Opiyo: **Prediction of precise subsoiling based on analytical method, discrete element simulation and experimental data from soil bin**, *Scientific Reports* (IF=3.8), 2021
- Qi, Chao, **Innocent Nyalala**, and Kunjie Chen: **Detecting the early flowering stage of tea chrysanthemum using the F-YOLO model**, *Agronomy* (IF=3.3), 2021
- **Innocent Nyalala**, Cedric Okinda, Chen Kunjie, Tchalla Korohou, Luke Nyalala, and Qi Chao: **Weight and volume estimation of poultry and products based on computer vision systems: a review**, *Poultry Science* (IF=3.8), 2021
- **Nyalala Innocent**, Cedric Okinda, Qi Chao, Peter Mecha, Tchalla Korohou, Zuo Yi, Samuel Nyalala, Zhang Jiayu, Liu Chao, and Chen Kunjie: **Weight and volume estimation of single and occluded tomatoes using machine vision**, *International Journal of Food Properties* (IF=3.1), 2021
- Yousaf, Khurram, Haolu Liu, Xu Gao, Chao Liu, Adnan Abbas, **Innocent Nyalala**, Muhammad Ahmad, Muhammad Ameen, and Kunjie Chen: **Influence of environmental conditions on drying efficiency and heat pump performance in closed and open loop drying of paddy**, *Drying Technology* (IF=2.7), 2020
- Q. Korohou, Tchalla, Okinda, C., **Nyalala, I.**, Torotwa, I., Mwangi, E., Wang, X., He, R., & Ding: **Separation of partially occluded rice kernels and variety classification based on computer vision**, *International Agricultural Engineering Journal* (IF=0.23), 2020
- Okinda, Cedric, Yuwen Sun, **Innocent Nyalala**, Tchalla Korohou, Samwel Opiyo, Jintao Wang, and Mingxia Shen: **Egg volume estimation based on image processing and computer vision**, *Journal of Food Engineering* (IF=5.3), 2020
- Korohou, Tchalla, Cedric Okinda, Haikang Li, Yifei Cao, **Innocent Nyalala**, Lianfei Huo, Mouloumdéma Potcho, Xiang Li, and Qishuo Ding: **Wheat grain yield estimation based on image morphological properties and wheat biomass**, *Journal of Sensors* (IF=1.4), 2020
- Okinda, Cedric, **Innocent Nyalala**, Tchalla Korohou, Celestine Okinda, Jintao Wang, Tracy Achieng, Patrick Wamalwa, Tai Mang, and Mingxia Shen: **A review on computer vision systems in monitoring of poultry: A welfare perspective**, *Artificial Intelligence in Agriculture* (IF=8.2), 2020.
- Okinda, Cedric, Mingzhou Lu, Longshen Liu, **Innocent Nyalala**, Caroline Muneri, Jintao Wang, Hailin Zhang, and Mingxia Shen: **A machine vision system for early detection and prediction of sick birds: A broiler chicken model**, *Biosystems Engineering* (IF=4.4), 2019
- **Nyalala Innocent**, Cedric Okinda, Luke Nyalala, Nelson Makange, Qi Chao, Liu Chao, Khurram Yousaf, and Kunjie Chen: **Tomato volume and mass estimation using computer vision and machine learning algorithms: Cherry tomato model**, *Journal of Food Engineering* (IF=5.3), 2019
- Yu, Zhenwei, Qi Gao, Yue Zhang, Dandan Wang, **Innocent Nyalala**, and Kunjie Chen: **Production of activated carbon from sludge and herb residue of traditional Chinese medicine industry and its application for methylene blue removal**, *BioResources* (IF=1.747), 2019
- Okinda, Cedric, Mingzhou Lu, **Innocent Nyalala**, Jiawei Li, and Mingxia Shen: **Asphyxia occurrence detection in sows during the farrowing phase by inter-birth interval evaluation**, *Computers and Electronics in Agriculture* (IF=7.7), 2018

- Okinda, C., MingXia Shen, JiaWei Li, **I. Nyalala**, LongShen Liu, MingZhou Lu, and GuangYue Zhang: **Detection of an onset of farrowing by classification of crated sow's activities**, *International Agricultural Engineering Journal* (IF=0.23), 2018
- Wang, Haoyun, Ke Wang, Duo Li, **Innocent Nyalala**, Yingjun Xiong, and Huanliang Xu: **Research on range-based localization algorithm with virtual force and time synchronization in wireless sensor and actor networks**, *International Journal of Communication Networks and Distributed Systems* (IF=1), 2016
- Okinda, Cedric Sean, **Innocent Nyalala**, Zhang Chi and Shen Mingxia: **Intelligent energy saving system**, *International Journal of Emerging Technology and Advanced Engineering* 2015

Academic Services

Member, MTech Project Committee: IIT Madras Zanzibar, 2024

Member, Faculty Recruitment Committee: IIT Madras Zanzibar, 2024

Reviewers for Journals: Journal of Electronics, MDPI 2024

Journal of AgriEngineering, MDPI 2024

Journal of Animals, MDPI 2024

Journal of Sensors, MDPI 2024

Computers and Electronics in Agriculture, Elsevier 2024

Journal of Food Engineering, Elsevier 2019-2023

Artificial Intelligence in Agriculture, Elsevier 2020-2022

Scientific Reports, Nature 2023

Head of Department: Department of Computer Science, The East African University. 2017

Senate Member: University Senate Member, The East African University. 2017

Teaching Experience

10/2024 -Present: Assistant Professor, *School of Engineering & Science*, IIT Madras Zanzibar, Zanzibar

04/2022-12/2024: Part-time Lecturer, *Department of Computer Science*, Egerton University, Nakuru

09/2022-12/2024: Part-time Lecturer, *Department of Literature, Language and Linguistics*, Egerton University

01/2023-04/2023: Part-time Lecturer, *Department of Accounting, Finance and Management*, Egerton University

02/2022-08/2022: Part-time Lecturer, *Department of Information Technology*, Maseno University, Yala

01/2017-08/2017: Part-time Tutorial Fellow, *Department of ICT & Engineering*, Zetech University, Nairobi

09/2016-08/2017: Tutorial Fellow, *Department of Computer Science*, The East African University, Kitengela

09/2016-12/2016: Part-time lecturer, *Department of Computer Science & Information Technology*, Kabarak University

Professional Memberships

Association for the Advancement of Artificial Intelligence (AAAI) 2024, Member ID: 638956

Association for Computing Machinery (ACM) 2024, Member ID: 5070064

Internet Society (ISOC) 2022, Kenya Chapter

International Association of Engineers (IAENG) 2024, Member ID: 259573

Leadership Experience

2013–2014: **President, International Students Association**, Nanjing Agricultural University, China

2009–2012: **Youth Chairperson** at Holy Trinity Catholic Parish Church, Nakuru, Kenya

China Scholarship Council (CSC), PhD Scholarship, Nanjing Agricultural University

References (with website links on names)

Name	Affiliation	Email address
Prof. Ramkrishna Pasumarthi	IIT Madras Zanzibar, Tanzania	ramkrishna@iitmz.ac.in
Prof. Nirav Bhatt	IIT Madras Zanzibar, Tanzania	niravbhatt@iitmz.ac.in
Prof. Chen Kunjie	Nanjing Agricultural University, China	kunjiechen@njau.edu.cn
Prof. Bai Zhentian	Nanjing Agricultural University, China	baizht@njau.edu.cn
Dr. Mary Manyange Bosire	Egerton University, Kenya	henrymary.bosire@egerton.ac.ke